

Key–Value Annotation of *Deep Residual Learning for Image Recognition*

He, Zhang, Ren & Sun — arXiv:1512.03385v1 [cs.CV], 10 Dec 2015

A hand annotation of the key–value pairs stated in the source paper: **547 distinct records** across **51 keys**. Values are read off the rendered PDF pages; the pipeline’s OCR of the same paper is deliberately not used as a source, so where the OCR is wrong this annotation still agrees with the paper.

Keys are Title Case, values are taken as printed, and the unit lives in the key rather than the value — the paper’s “a depth of up to 152 layers” is recorded as **Depth** / 152. Each distinct (*key*, *value*) pair is recorded **once for the whole paper**, so the count is a count of distinct facts rather than of sentences. The **Page** column is annotation metadata, not part of the pair: it records where the pair was first seen.

Conventions, and the decisions still open — chiefly that this flat *key:value* form cannot express which table row a number belongs to — are written up in **ANNOTATION.md**.

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1 Bibliographic

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Table 2: Author

Key	Value	Page
Author	Kaiming He	p.1
Author	Xiangyu Zhang	p.1
Author	Shaoqing Ren	p.1
Author	Jian Sun	p.1

Table 3: Organization

Key	Value	Page
Organization	Microsoft Research	p.1

Table 4: Email

Key	Value	Page
Email	{kahe, v-xiangz, v-shren, jiansun}@microsoft.com	p.1

Table 5: Identifier

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Identifier	arXiv:1512.03385v1	p.1

Table 6: Category

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Table 7: Date

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2 Datasets and tasks

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Size	50k validation images	p.4
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Size	50k training images	p.7
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3 Architecture

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4 Method

Table 24: Method

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Concept	residual mapping	p.3
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Learning rate	0.001	p.10
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Batch size	16 images	p.10

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Iterations	400 iterations	p.7
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Iterations	80k	p.10

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6 Results

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Error	27.88	p.5
Error	28.54	p.5
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Error	28.07	p.6
Error	9.33	p.6
Error	9.15	p.6
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Error	7.40	p.6
Error	22.85	p.6
Error	6.71	p.6
Error	21.75	p.6
Error	6.05	p.6
Error	21.43	p.6
Error	5.71	p.6
Error	8.43	p.6
Error	7.89	p.6
Error	24.4	p.6
Error	7.1	p.6
Error	21.59	p.6
Error	21.99	p.6
Error	5.81	p.6
Error	21.84	p.6
Error	21.53	p.6
Error	5.60	p.6
Error	20.74	p.6
Error	5.25	p.6
Error	19.87	p.6
Error	4.60	p.6
Error	19.38	p.6
Error	4.49	p.6
Error	7.32	p.6
Error	6.66	p.6
Error	6.8	p.6
Error	4.94	p.6
Error	4.82	p.6
Error	3.57	p.6
Error	9.38	p.7
Error	8.81	p.7
Error	8.22	p.7
Error	8.39	p.7
Error	7.54 (7.72±0.16)	p.7
Error	8.80	p.7
Error	8.75	p.7
Error	7.51	p.7
Error	7.17	p.7
Error	6.97	p.7
Error	6.43 (6.61±0.16)	p.7
Error	7.93	p.7
Error	4.49%	p.7
Error	6.43%	p.8

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Error	0.1%	p.8
Error	33.1%	p.12
Error	13.3%	p.12
Error	11.7%	p.12
Error	14.4%	p.12
Error	10.6%	p.12
Error	9.0%	p.12
Error	4.6%	p.12
Error	33.1	p.12
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mAP	76.4	p.8
mAP	73.8	p.8
mAP	41.5	p.8
mAP	21.2	p.8
mAP	48.4	p.8
mAP	27.2	p.8
mAP	49.9	p.11
mAP	29.9	p.11
mAP	51.1	p.11
mAP	30.0	p.11
mAP	53.3	p.11
mAP	32.2	p.11
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mAP	34.9	p.11
mAP	59.0	p.11
mAP	37.4	p.11
mAP	43.9	p.11
mAP	60.5	p.11
mAP	58.8	p.11
mAP	63.6	p.11
mAP	62.1	p.11
mAP	85.6	p.11
mAP	83.8	p.11

Table 46: AP

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AP	79.0	p.11
AP	70.9	p.11
AP	65.5	p.11
AP	52.1	p.11
AP	83.1	p.11
AP	84.7	p.11
AP	86.4	p.11
AP	52.0	p.11
AP	81.9	p.11
AP	65.7	p.11
AP	84.8	p.11
AP	84.6	p.11
AP	77.5	p.11
AP	76.7	p.11
AP	38.8	p.11
AP	73.6	p.11
AP	73.9	p.11
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AP	72.6	p.11
AP	79.8	p.11
AP	80.7	p.11
AP	76.2	p.11
AP	68.3	p.11
AP	55.9	p.11
AP	85.1	p.11
AP	85.3	p.11

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AP	89.8	p.11
AP	56.7	p.11
AP	87.8	p.11
AP	69.4	p.11
AP	88.3	p.11
AP	88.9	p.11
AP	80.9	p.11
AP	78.4	p.11
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AP	78.6	p.11
AP	72.0	p.11
AP	90.0	p.11
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AP	76.1	p.11
AP	89.9	p.11
AP	75.5	p.11
AP	90.3	p.11
AP	89.1	p.11
AP	88.7	p.11
AP	65.4	p.11
AP	88.1	p.11
AP	85.6	p.11
AP	89.0	p.11
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AP	74.3	p.11
AP	53.9	p.11
AP	49.8	p.11
AP	75.9	p.11
AP	88.5	p.11
AP	45.6	p.11
AP	77.1	p.11
AP	55.3	p.11
AP	86.9	p.11
AP	81.7	p.11
AP	79.6	p.11
AP	40.1	p.11
AP	60.9	p.11
AP	81.2	p.11
AP	61.5	p.11
AP	86.5	p.11
AP	81.6	p.11
AP	77.2	p.11

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Table 46: AP (*continued*)

Key	Value	Page
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AP	51.0	p.11
AP	76.6	p.11
AP	93.2	p.11
AP	48.6	p.11
AP	80.4	p.11
AP	59.0	p.11
AP	92.1	p.11
AP	48.1	p.11
AP	77.3	p.11
AP	66.5	p.11
AP	65.6	p.11
AP	88.4	p.11
AP	71.4	p.11
AP	86.3	p.11
AP	94.2	p.11
AP	66.8	p.11
AP	89.4	p.11
AP	69.2	p.11
AP	93.9	p.11
AP	91.9	p.11
AP	90.9	p.11
AP	67.9	p.11
AP	88.2	p.11
AP	76.8	p.11
AP	80.0	p.11

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Class	bike	p.11
Class	bird	p.11
Class	boat	p.11
Class	bottle	p.11
Class	bus	p.11
Class	car	p.11
Class	cat	p.11
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Class	cow	p.11
Class	table	p.11
Class	dog	p.11
Class	horse	p.11

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Table 47: Class (*continued*)

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Class	mbike	p.11
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Class	plant	p.11
Class	sheep	p.11
Class	sofa	p.11
Class	train	p.11
Class	tv	p.11

Table 48: Improvement

Key	Value	Page
Improvement	28%	p.1
Improvement	18%	p.3
Improvement	2.8%	p.5
Improvement	3.5%	p.6
Improvement	6.0%	p.8
Improvement	6%	p.10
Improvement	6.9%	p.10
Improvement	>3%	p.10
Improvement	about 2 points	p.10
Improvement	about 1 point	p.10
Improvement	over 2 points	p.11
Improvement	10 points	p.11
Improvement	8.5 points	p.12
Improvement	64%	p.12

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Key	Value	Page
Comparison	8x	p.1

Table 50: Competition

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Competition	ILSVRC 2015	p.1
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Table 51: Placement

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Placement	1st places	p.1
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